GITAM Deemed to be University

Aerospace Engineering | Course Code: AE301

Aerodynamics & Flight Mechanics - Aerodynamics Theory, Lift Generation & Fl

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Overview: Kutta-Joukowski theorem, induced drag calculation, supersonic oblique shock relations

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Fundamental Incompressible Potential Flow
Continuity and Euler equations, Velocity potential, Stream function, Elementary flows Source, Sink,

Unit 2: Incompressible Airfoil & Finite Wing Theory
Kutta condition, Thin airfoil theory for symmetric and cambered airfoils, Prandtl lifting line theor

Unit 3: Compressible Flow & Shock Waves
Speed of sound, Mach number regimes, 1D isentropic flow through converging-diverging nozzles, Normal

Unit 4: Aircraft Performance & Flight Stability
Thrust required and power required curves, Rate of climb, Service ceiling, Static longitudinal stabi

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the Kutta-Joukowski theorem L = rho * V_inf * Gamma for lift generation on a spinning cylinder
2. Using Thin Airfoil Theory, prove that the lift curve slope dC_L/dalpha equals 2*pi for a thin symmetri
3. Formulate the normal shock relations for static pressure ratio P2/P1 and downstream Mach number M2 in

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* John D. Anderson Jr. - Fundamentals of Aerodynamics McGraw-Hill 6th Ed
* E.L. Houghton, P.W. Carpenter - Aerodynamics for Engineering Students Butterworth-Heinemann

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